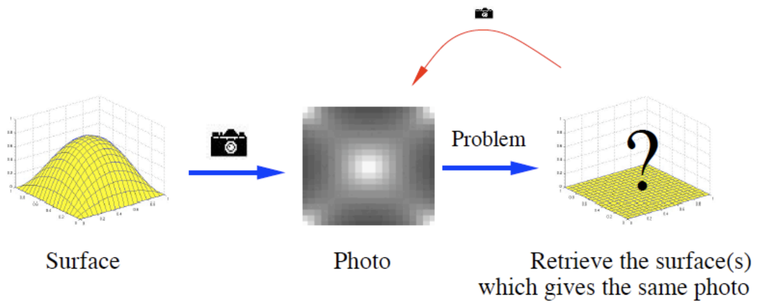
Direct Shape Recovery from Photometric Stereo with Shadows[a]

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[a]This research was partly supported by European Community's FP7- ERC program, grant agreement no. 267414 and by Broadcom foundation.

Shape from Shading problem



Main idea

Reconstruction of the shape of an object starting from a digital image of the object.

Assumptions

1. the surface is Lambertian;
2. the light source is placed at infinity;
3. shadows or missing image parts occur;
4. the albedo is unknown;
5. there are no self-reflections on the surface.

Problem formulation

Find the solution(s) $z(x,y)$ of the irradiance equation

$$\rho(x,y)[\omega \cdot n(x,y)] = I(x,y)$$

through the non-linear partial differential equation associated

$$\rho(x,y)\frac{-z_x(x,y)\omega_1 - z_y(x,y)\omega_2 + \omega_3}{\sqrt{1+|\nabla z(x,y)|^2}} = I(x,y), \quad \text{for} \quad (x,y) \in \Omega$$

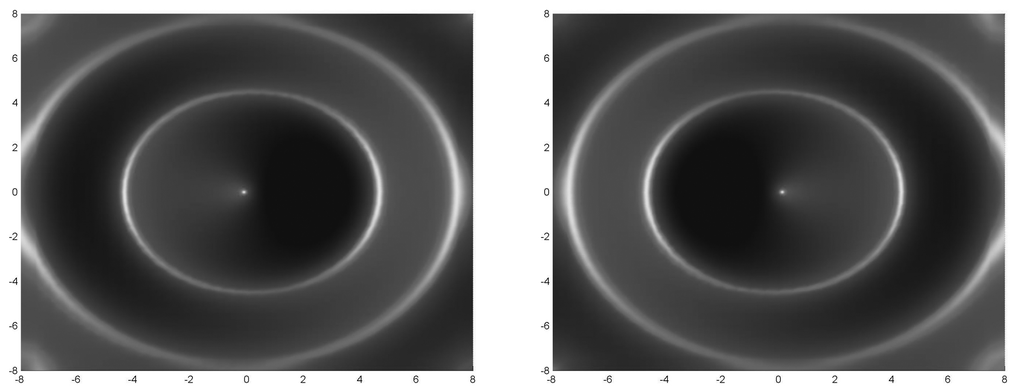
where:

- $\rho(x,y)$ is the albedo;
- ω is the unit vector for the light source;
- $n(x,y)$ is the unit normal vector outgoing to the surface in the perspective world;
- $I(x,y)$ is the grayscale image;
- Ω is the image domain.

Drawback:

There is no unique solution.

Shape from Shading - Photometric Stereo



Increase the number of digital images: *Photometric Stereo*

The photometric stereo technique consists in adding information to the Shape from Shading problem using more than one image such that:

- the point of view is the same;
- the light source changes for each image (ω', ω'', ω''', ...).

Differential problem: system of Hamilton-Jacobi PDEs

$$\begin{cases} \rho(x,y)\dfrac{-z_x(x,y)\omega_1' - z_y(x,y)\omega_2' + \omega_3'}{\sqrt{1+|\nabla z(x,y)|^2}} = I_1(x,y), & \text{on } \Omega \\ \rho(x,y)\dfrac{-z_x(x,y)\omega_1'' - z_y(x,y)\omega_2'' + \omega_3''}{\sqrt{1+|\nabla z(x,y)|^2}} = I_2(x,y), & \text{on } \Omega \\ z(x,y) = g(x,y) & \text{on } \partial\Omega. \end{cases} \tag{1}$$

Existing Result

In [1] it is proved uniqueness of weak (Lipschitz) solution for the following linear differential formulation with Dirichelet boundary condition:

$$\begin{cases} b(x,y)\cdot\nabla z(x,y) = f(x,y), \ a.e.(x,y) \in \Omega^p \\ z(x,y) = g(x,y) \qquad\qquad \forall (x,y) \in \partial\Omega \end{cases} \tag{2}$$

where

$$b(x,y) = (I_2(x,y)\omega_1' - I_1(x,y)\omega_1'', I_2(x,y)\omega_2' - I_1(x,y)\omega_2'')$$

and

$$f(x,y) = I_2(x,y)\omega_3' - I_1(x,y)\omega_3''.$$

where the images are discontinuous functions with no shadows.

Multiple Images Formulation

If we consider the problem (2) for three images, we get the following three linear equations (one for each couple of images):

$$\begin{cases} b^{(1,2)}(x,y)\cdot\nabla z(x,y) = f^{(1,2)}(x,y), & \text{a.e. } (x,y)\in\Omega \\ b^{(1,3)}(x,y)\cdot\nabla z(x,y) = f^{(1,3)}(x,y), & \text{a.e. } (x,y)\in\Omega \\ b^{(2,3)}(x,y)\cdot\nabla z(x,y) = f^{(2,3)}(x,y), & \text{a.e. } (x,y)\in\Omega \end{cases} \tag{3}$$

Exploiting the linearity of the previous PDEs, by summing the three equations, we can resume the previous system in the following equation:

$$\begin{cases} b_3(x,y)\cdot\nabla z(x,y) = f_3(x,y), \ \text{a.e. } (x,y)\in\Omega \\ z(x,y) = g(x,y) \qquad\qquad \forall (x,y)\in\partial\Omega^p \end{cases} \tag{4}$$

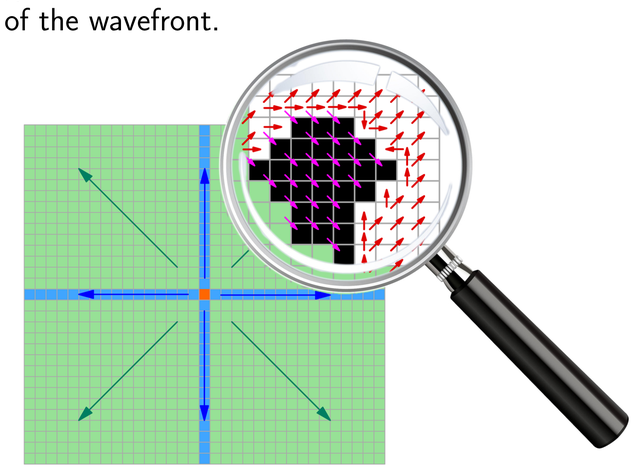
where

$$b_3(x,y) = \alpha(x,y)b^{(1,2)}(x,y) + \beta(x,y)b^{(1,3)}(x,y) + \gamma(x,y)b^{(2,3)}(x,y) \tag{5}$$

$$f_3(x,y) = \alpha(x,y)f^{(1,2)}(x,y) + \beta(x,y)f^{(1,3)}(x,y) + \gamma(x,y)f^{(2,3)}(x,y). \tag{6}$$

Integration Strategy with Shadows

We fix an arbitrary point (in orange) and perform surface recovery along an expanding wavefront of visited pixels. When shadows/occlusions occur, we orient the vector field (i.e. we choose the coefficients α, β and γ) in order to avoid them when their directions are inconsistent with the direction of expansion of the wavefront.



Numerical Schemes

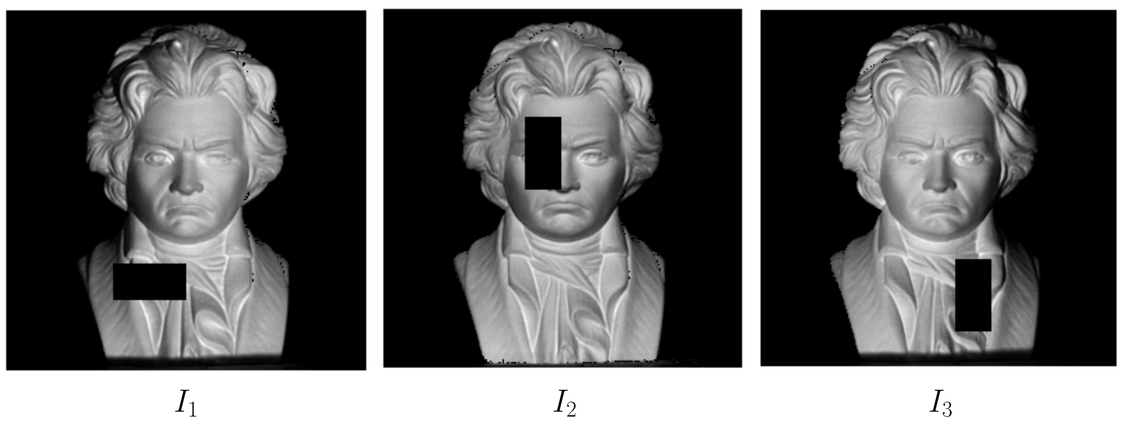
Dividing both sides of the equation in (2) by $||b(x,y)||$, considering the definition of the directional derivative we arrive to the following fixed point semi-Lagrangian schemes:

$$\text{(sL)} \quad Z_{i,j}^{t+1} = Z^t(x_i - hr_{i,j}^1, y_j - hr_{i,j}^2) + \frac{f_{i,j}}{|b_{i,j}|}h \tag{7}$$

where $r_{i,j} = \frac{b_{i,j}}{||b_{i,j}||}$ and $Z^{t+1}(x_i,y_j) = Z_{i,j}^{t+1}$ is the approximation of $z(x_i,y_j)$ on a uniform discretization of Ω (Ω_d).

Numerical Tests

Real images



I_1 I_2 I_3

Figure : Images of the Beethoven bust with occlusions.

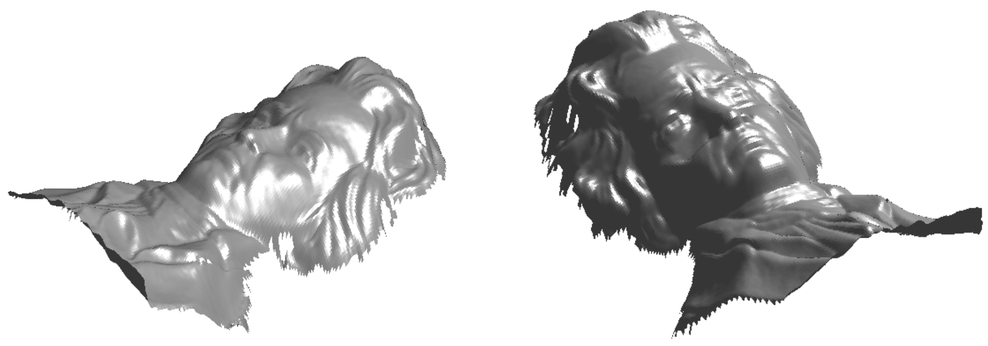


Figure : Two different perspectives of the reconstructed bust of Beethoven.

Conclusion and Perspective

We presented a new method for the shape recovery solving the Shape from Photometric Stereo problem. The approach is based on a new differential method where the reconstruction of the surface is direct and does not need of the double step procedure based on the computation of the normal to the surface before the surface reconstruction. This new formulation allows to recover the surface even if shadows or occlusions occur. Our current implementation could potentially be accelerated by transferring it into a streaming programming environment such as CUDA to implement a fully parallelized version of the suggested integration strategy on a GPU.

References

[1] R. Mecca and M. Falcone, *Uniqueness and approximation of a Photometric Shape-from-Shading model*, SIAM Journal on Imaging Sciences, 6(1):616–659, 2013.

[2] K. Goebel and W. A. Kirk, *A Fixed Point Theorem for Asymptotically Nonexpansive Mappings*, Proceedings of the American Mathematical Society, vol. 35, no. 1, pp. 171-174, 1972

[3] R. Mecca, A. Tankus and A. M. Bruckstein *Two-Image Perspective Photometric Stereo Using Shape-from-Shading*, ACCV (4) 2012: 110-121